

$$1) F_D = f(\rho, \mu, d, z, V)$$

$$n=6$$

Expressing in terms of base dimensions:

$$F_D = ma = MLT^{-2}$$

$$\rho = ML^{-3}$$

$$\mu = ML^{-1}T^{-1}$$

$$d = L$$

$$z = L$$

$$V = LT^{-1}$$

$$F_D \rho^a V^b d^c = \overset{\checkmark}{M} \overset{\checkmark}{L} \overset{\checkmark}{T}^{-2} \overset{\checkmark}{M}^a \overset{\checkmark}{L}^{-3a} \overset{\checkmark}{L}^b \overset{\checkmark}{T}^{-b} \overset{\checkmark}{L}^c$$

$$= M^{a+1} L^{b+1-3a+c} T^{-2-b}$$

$$a+1=0$$

$$b+1-3a+c=0$$

$$-2-b=0$$

$$a=-1$$

$$b=-2$$

$$c=-2$$

$$\therefore \pi_1 = \frac{F_D}{\rho V^2 d^2}$$

$$\begin{aligned}
 2) \mu \rho^a v^b d^c &= \check{M} \check{L}^{-1} \check{T}^{-1} \check{M}^a \check{L}^{-3a} \check{L}^b \check{T}^{-b} \check{L}^c \\
 &= M^a L^{b+c-3a} T^{-b} M L^{-1} T^{-1} \\
 &= M^{a+1} L^{b+c-3a-1} T^{-b-1}
 \end{aligned}$$

$$a+1=0 \rightarrow a=-1$$

$$b+c-3a-1=0 \rightarrow -3a+b+c=1$$

$$-b-1=0 \rightarrow b=-1$$

$$\therefore a=-1$$

$$b=-1$$

$$c=-1$$

$$\therefore \pi_2 = \frac{\mu}{\rho v d}$$

$$\begin{aligned}
 3) 2 \rho^a v^b d^c &= M^a L^{b+c-3a} T^{-b} L \\
 &= M^a L^{b+c-3a+1} T^{-b}
 \end{aligned}$$

$$a=0$$

$$b+c-3a+1=0 \rightarrow -3a+b+c=-1$$

$$-b=0 \rightarrow b=0$$

$$\therefore a=0, b=0, c=-1$$

$$\pi_3 = \frac{2}{d}$$

$$\therefore \frac{F_D}{\rho d^2 v^2} = f\left(\frac{\mu}{\rho v d}, \frac{2}{d}\right)$$

$$2) \Delta p = f(D, \omega, \rho, Q)$$

$$n = 5$$

Expressing everything in base dimensions:

$$\Delta p = M L^{-1} T^{-2}$$

$$D = L$$

$$\omega = T^{-1}$$

$$\rho = M L^{-3}$$

$$Q = L^3 T^{-1}$$

ρ, ω, D are the repeating variables

$$\begin{aligned} \Delta p \rho^a \omega^b D^c &= M^a L^{-3a} T^{-b} L^c M L^{-1} T^{-2} \\ &= M^a L^{c-3a} T^{-b} M L^{-2} T^{-2} \\ &= M^{a+1} L^{c-3a-1} T^{-b-2} \end{aligned}$$

$$a+1=0 \rightarrow a=-1$$

$$c-3a-2=0 \rightarrow -3a+c=1$$

$$-b-2=0 \rightarrow -b=2$$

Solving:

$$a = -1$$

$$b = -2$$

$$c = -2$$

$$\pi_1 = \frac{\Delta p}{\rho \omega^2 D^2}$$

$$2) Q \rho^a \omega^b D^c = M^a L^{c-3a} T^{-b} L^3 T^{-1} \\ = M^a L^{c-3a+3} T^{-b-1}$$

$$a=0$$

$$c-3a+3=0 \rightarrow -3a+c=-3$$

$$-b-1=0 \rightarrow -b=1$$

Solving:

$$a=0$$

$$b=-1$$

$$c=-3$$

$$\pi_2 = \frac{Q}{\omega D^3}$$

$$\therefore \frac{\Delta p}{\rho \omega^2 D^2} = f\left(\frac{Q}{\omega D^3}\right)$$

$$3a) \frac{\Delta p}{2} = f(Q, D, \mu)$$

Expressing everything in basic dimensions:

$$\Delta p = ML^{-1}T^{-2} \Rightarrow \frac{\Delta p}{2} = ML^{-2}T^{-2}$$

$$L = L$$

$$Q = L^3T^{-1}$$

$$D = L$$

$$\mu = ML^{-1}T^{-1}$$

Assuming the repeating terms to be Q, D, μ .

$$\begin{aligned} \frac{\Delta p}{2} Q^a D^b \mu^c &= L^{\overset{\checkmark}{3a} \overset{\checkmark}{T^{-a}} L^{\overset{\checkmark}{b}} M^{\overset{\checkmark}{c}} L^{\overset{\checkmark}{-c}} T^{\overset{\checkmark}{-c}} M L^{\overset{\checkmark}{-2}} T^{\overset{\checkmark}{-2}}} \\ &= L^{3a+b-c-2} T^{-a-c-2} M^{c+1} \end{aligned}$$

$$3a+b-c=2$$

$$-a - c = 2$$

$$c = -1$$

Solving:

$$a = -1$$

$$b = 4$$

$$c = -1$$

$$\pi_1 = \frac{\Delta p D^4}{2 Q \mu}$$

$$\text{Constant } C = \frac{D^4 \Delta p}{\mu 2 Q}$$

$$3b) \quad C = \frac{D^4 \Delta p}{\mu L Q}$$

$$C = \frac{D^4}{\mu L} \frac{\Delta p}{Q}$$

From the table,

$$C_1 = \frac{(2 \times 10^{-3})^4}{0.004 \times 300 \times 10^{-3}} \left(\frac{1.1 \times 10^4}{3.6 \times 10^{-6}} \right)$$

$$= \frac{4}{3} \times 10^{-8} (3.055 \times 10^9)$$

$$= \frac{1100}{27}$$

$$C_2 = \frac{4}{3} \times 10^{-8} \left(\frac{1.5 \times 10^4}{4.9 \times 10^{-6}} \right)$$

$$= \frac{2000}{49}$$

$$C_3 = \frac{4}{3} \times 10^{-8} \left(\frac{1.9 \times 10^4}{6.3 \times 10^{-6}} \right)$$

$$= \frac{7600}{189}$$

$$C_4 = \frac{4}{3} \times 10^{-8} \left(\frac{2.4 \times 10^4}{7.9 \times 10^{-6}} \right)$$

$$= \frac{3200}{79}$$

$$C_5 = \frac{4}{3} \times 10^{-8} \left(\frac{3.0 \times 10^4}{9.8 \times 10^{-6}} \right)$$

$$= \frac{2000}{49}$$

$$C_{avg} = \frac{1}{5} \left(\frac{1100}{27} + \frac{2000}{49} + \frac{7600}{189} + \frac{3200}{79} + \frac{2000}{49} \right) = 40.61827263 \approx 40.618$$

$$4a) \theta = f(\omega, \mu, K, D, D_2, l)$$

Expressing variables in basic dimensions:

$$\theta = 0$$

$$\omega = T^{-1}$$

$$\mu = ML^{-1}T^{-1} \rightarrow \text{repeating}$$

$$K = ML^2T^{-2} \rightarrow \text{repeating}$$

$$D_1 = L$$

$$D_2 = L$$

$$l = L \rightarrow \text{repeating}$$

$$\pi_1 = \theta \because \theta \text{ is dimensionless}$$

$$\omega \mu^a K^b l^c = T^{-1} \overset{\checkmark}{M}^a \overset{\checkmark}{L}^{-a} \overset{\checkmark}{T}^{-a} \overset{\checkmark}{M}^b \overset{\checkmark}{L}^{2b} \overset{\checkmark}{T}^{-2b} \overset{\checkmark}{L}^c$$

$$= M^{a+b} L^{2b+c-a} T^{-a-2b-1}$$

$$= M^{a+b} L^{-a+2b+c} T^{-a-2b-1}$$

$$a+b=0$$

$$-a+2b+c=0$$

$$-a-2b=1$$

Solving:

$$a=1$$

$$b=-1$$

$$c=3$$

$$\therefore \pi_2 = \frac{\omega \mu l^3}{K}$$

4a) Unit of $D_1 = D_2$, so we just solve for D .

$$D M^a K^b L^c = M^{a+b} L^{-a+2b+c} T^{-a-2b} L$$
$$= M^{a+b} L^{-a+2b+c+1} T^{-a-2b}$$

$$a+b=0$$

$$-a+2b+c=-1$$

$$-a-2b=0$$

$$a=0$$

$$b=0$$

$$c=-1$$

$$\therefore \pi_3 = \frac{D_1}{L} \quad \pi_4 = \frac{D_2}{L}$$

$$\therefore \theta = f\left(\frac{\omega \mu l^3}{K}, \frac{D_1}{L}, \frac{D_2}{L}\right)$$

Since D_1 and D_2 are constants,

$$\theta = f\left(\frac{\omega \mu l^3}{K}\right)$$

$$4b) \theta = f\left(\frac{\mu l^3}{K} \omega\right)$$

$$\frac{\mu l^3}{K} = \frac{0.5(0.3)^3}{14}$$

$$= \frac{27}{28000}$$

$\theta(\text{rad})$	$\omega(\text{rad s}^{-1})$	$\frac{\omega \mu l^3}{K}$
0.89	0.30	$\frac{81}{280000}$
1.50	0.50	$\frac{27}{56000}$
2.51	0.82	$7.907142857 \times 10^{-4}$
3.05	1.05	1.0125×10^{-3}
4.28	1.43	$1.378928571 \times 10^{-3}$
5.52	1.86	$1.793571429 \times 10^{-3}$
6.40	2.14	$2.063571429 \times 10^{-3}$

The graph is linear, hence

$$m = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \frac{6.40 - 0.89}{(2.14 - 0.30) \frac{27}{28000}}$$

$$= 3105.47504$$

$$\theta = 3105.47504 \left(\frac{\omega \mu l^3}{K} \right)$$

$$\begin{aligned}
 4b) \quad \theta &= 3105.47504 \left(\frac{\omega \mu l^3}{k} \right) \\
 &= 3105.47504 \left(\frac{\omega \mu (0.3)^3}{14} \right) \\
 &= \frac{551}{92} \omega \mu \\
 &\approx 5.99 \omega \mu
 \end{aligned}$$